

Pandu Ranga Reddy Konala

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Personal Statement

I'm a cybersecurity researcher working at the crossroads of AI code generation and modern DevSecOps. My PhD research focuses on catching vulnerabilities in Infrastructure as Code ranging from misconfigurations to supply chain attacks by tracing how security smells spread through metadata patterns. More recently, I've been building agentic pipelines that generate, validate, and self-repair security-hardened quality infrastructure against regulatory frameworks which reduces manual compliance effort from weeks to under an hour and designed distillation pipelines to bring these capabilities to smaller, cost-efficient open-source AI models.

Academic Progress

The University of Waikato

PhD in Computer Science

- Fully funded by MBIE, New Zealand, through the AI for Human-Centric Security Catalyst project scholarship
- Thesis: *A Code Quality Framework for Infrastructure as Code in the Modern DevSecOps Era*
- Published 3 first-author papers at peer-reviewed venues (IEEE CSR, DBSec, IWSEC) with 2 under review

Hamilton, New Zealand

April 2022 - February 2026

Lancaster University

Masters in Cyber Security

- Fully funded by Lancaster University through the Cyber Security Scholarship program
- NCSC-certified programme by the National Cyber Security Centre, UK

Lancaster, United Kingdom

September 2019 - September 2020

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science and Engineering

- Specialized in Cyber Security
- Published 2 first-author research papers on cloud security and decentralized file systems

Coimbatore, India

June 2015 - April 2019

Work Experience

Research Associate

School of Computing & Mathematical Sciences, The University of Waikato

- Built an agentic quality-compliance pipeline that generates, validates, and self-repairs security-hardened Infrastructure as Code against regulatory frameworks like HIPAA, PCI DSS, and SOC 2 while achieving 100% compliance across CIS technical controls.
- Developed a knowledge-base-driven architecture with agentic LLM repair loops, automated validation, and containerized testing infrastructure, alongside a teacher-student distillation pipeline for SFT of smaller open-source models using a custom reward function for quality and compliance-aware IaC generation.

Hamilton, New Zealand

February 2021 - February 2026

Software Developer

Quantum Base

- Designed and implemented secure system architectures to integrate quantum PUFs with IoT devices, bridging the gap between quantum technology and everyday applications.

Lancaster, UK

June 2020 - September 2020

Security Analyst Intern

Infosec Future Pvt Ltd

- Conducted website penetration testing, vulnerability scanning, and risk analysis using automated tools and manual techniques during an eight-week internship.
- Performed detailed information gathering to identify security risks and vulnerabilities.

Lucknow, India

May 2018 - June 2018

Red Hat System Administrator Intern

Complete Open Source Solutions (COSS)

- Gained hands-on experience in managing and maintaining Red Hat Enterprise Linux systems during a four-week internship.
- Handled system installation, configuration, performance monitoring, security management and maintained system documentation.

Hyderabad, India

May 2017 - June 2017

Ethical Hacker Intern

Star Computers

- Gained hands-on experience in penetration testing and vulnerability assessments over a four-week period.
- Identified potential vulnerabilities and security risks in client systems and applications through various tests and assessments.

Vizag, India

May 2016 - May 2016

Professional Certifications

Red Hat Certified Engineer (RHCE)

by Red Hat, License No. 170-138-911

Red Hat Certified System Administrator (RHCSA)

by Red Hat, License No. 170-138-911

Certified Ethical Hacker (CEH)

by EC-Council, License No. ECC022209621222

Skills

Cyber Security Tools

Metasploit Framework, Nessus, Burp Suite, Wireshark, CIS-CAT

Programming

Python, Bash/Shell, Java, SQL, Node.js, R

Platforms & Infrastructure

AWS (EC2), Red Hat Linux, Kali, Docker, Podman, Ansible, Terraform

Miscellaneous

LaTeX, Git, CI/CD Pipelines, Static Analysis, LLM/GenAI Integration, Hack The Box

Peer-Reviewed Research Publications

Tracking Security Smell Diffusion Patterns in Ansible Playbooks Using Metadata

Research Paper (First-Author)

Published
November 2025

- International Workshop on Security (IWSEC) 2025, Japan

Metadata Assisted Supply-Chain Attack Detection for Ansible

Research Paper (First-Author)

Published
June 2025

- Conference on Data and Applications Security and Privacy (DBSec), Norway

SoK: Static Configuration Analysis in Infrastructure as Code Scripts

Research Paper (First-Author)

Published
August 2023

- IEEE International Conference on Cyber Security and Resilience (CSR), Venice, Italy

Access mechanism using inter planetary file system

Research Paper (First-Author)

Published
April 2019

- International Journal of Engineering and Technology, Volume 7, No 4, ISSN: 2227-524X

Securing Data in Cloud – A Physical Cyber System

Research Paper (First-Author)

Published
December 2017

- Journal of Computer Science Engineering, Volume-3, Issue-12, ISSN-2456-1843

A Policy-Driven Framework for Measuring the Code Quality of Ansible Roles: An Empirical Study

Research Paper (First-Author)

Under Review
2026

- Journal of Systems and Software (JSS)

Evaluating and Preventing Security Smells in AI-Generated Ansible Code

Research Paper (First-Author)

Under Review
2026

Projects

AI for Human Centric Security | MBIE Catalyst Project (in partnership with CSIRO Data61)

PhD Project

The University of Waikato

April 2022 - February 2026

- Part of a trans-Tasman cyber security research collaboration between New Zealand universities and CSIRO Data61, Australia, funded by MBIE.
- Developed a policy-driven code quality framework for Infrastructure as Code, validated across 26,000+ open-source repositories, detecting misconfigurations, supply chain attacks, and security smell diffusion patterns.
- Extended the framework to evaluate and fix code quality-compliance issues in AI-generated IaC code from LLMs.
- Built an end-to-end automated quality-compliance pipeline with multi-turn LLM repair, Docker/EC2 integration, and a web-based reporting interface for real-time pipeline monitoring and compliance visualization.

Cryptographically Secure On-Line Identity System

Masters Project

Lancaster University

June 2020 - September 2020

- Developed a framework leveraging quantum computer-generated identities (PUFs) for device authentication and validation, targeting resource-constrained IoT and mobile devices. Funded by Quantum Base and The Royal Society, UK.

Secure communication using IPFS with IoT

Under Graduate Project

Amrita Vishwa Vidyapeetham

June 2018 - April 2019

- Used the Interplanetary File System (IPFS) protocol for secure, direct communication between users and IoT devices, eliminating the need for third-party vendors.

Voluntary Activities

New Zealand Cyber Security Challenge

cybersecuritychallenge.org.nz

2022 - 2025

- Organized and facilitated Capture The Flag (CTF) events for the New Zealand Cyber Security Challenge, engaging over 500 participants, including high school students, university students, and industry professionals, to enhance their cybersecurity skills and awareness.

Cyber Security Workshop

Amrita School of Engineering, India

February 2016

- Organized and led a two-day workshop on Ethical Hacking and Cyber Security at Amrita School of Engineering, India, aimed at inspiring students from underprivileged backgrounds to explore opportunities in cybersecurity.

Interests

Hack The Box

I enjoy breaking into things (legally), active on HTB challenges for penetration testing practice.

Swimming

Competitive swimmer since I was 10, still drawn to anything involving water and breath control.

Travel

Love exploring new places, trying local food, and getting lost in history along the way.

Video Games

Big fan of story-driven games, if it has a good narrative, I'm in.